AP PRECALCULUS

Unit 1 • Lesson 1.6 practice

Lesson 1.6 • Polynomial Functions and End Behavior

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to \infty} \left(-4x^3 + x^2\right)$ is

(A) ∞

(B) $-\infty$

(C) -4

(D) 0

Question 2

NO CALCULATOR

Let $p(x) = (2 - x)^3$. What is $\lim_{x \to -\infty} p(x)$?

(A) $-\infty$

(B) $+\infty$

(C) 8

(D) 0

Question 3

NO CALCULATOR

Which polynomial increases without bound as $x \to \infty$ and as $x \to -\infty$?

(A) $x^3 + 1$

(B) $-x^4$

(C) $5x - x^2$

(D) $2x^6 - x$

Question 4

NO CALCULATOR

The graph of a polynomial falls to the left and rises to the right. Its degree and leading coefficient are

(A) even, positive

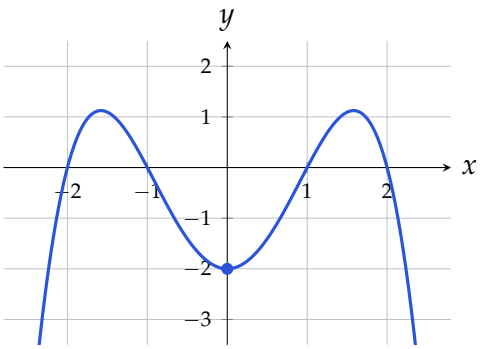
(B) even, negative

(C) odd, positive

(D) odd, negative

Question 5 (graph) NO CALCULATOR

The graph of a polynomial r is shown. Its zeros are ± 1 and ± 2, and $r(0) = -2$. Which could be $r(x)$?



(A) $x^4 - 5x^2 + 4$
(B) $-x^3 + 4x$
(C) $-\frac{1}{2}\left(x^4 - 5x^2 + 4\right)$
(D) $\frac{1}{2}\left(x^4 - 5x^2 + 4\right)$

Question 6 (Free response) NO CALCULATOR

Let $f(x) = -2x^4 + 3x^3 + x - 7$ and $g(x) = x^2(3 - x)(x + 1)^2$.

(a) Write limit statements for the end behavior of f.
(b) Find the degree and leading term of g, and write limit statements for its end behavior.
(c) Could the graph of g have 5 turning points? Explain.